

LECTURE-5

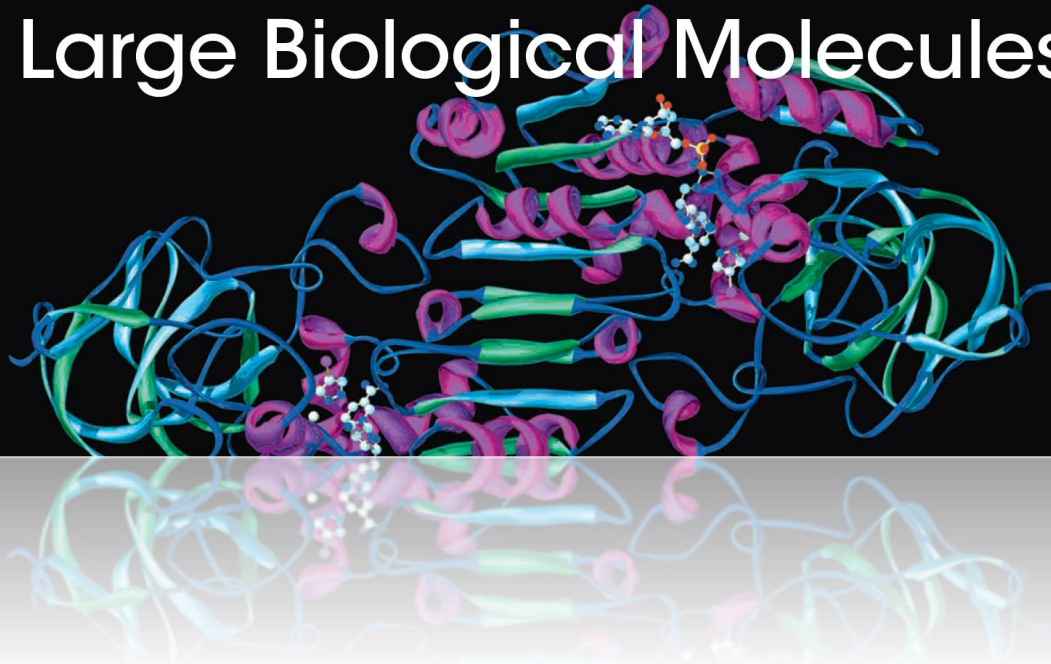
Macromolecules



INDRAPRASTHA INSTITUTE *of*
INFORMATION TECHNOLOGY
DELHI

5

The Structure and Function of Large Biological Molecules



KEY CONCEPTS

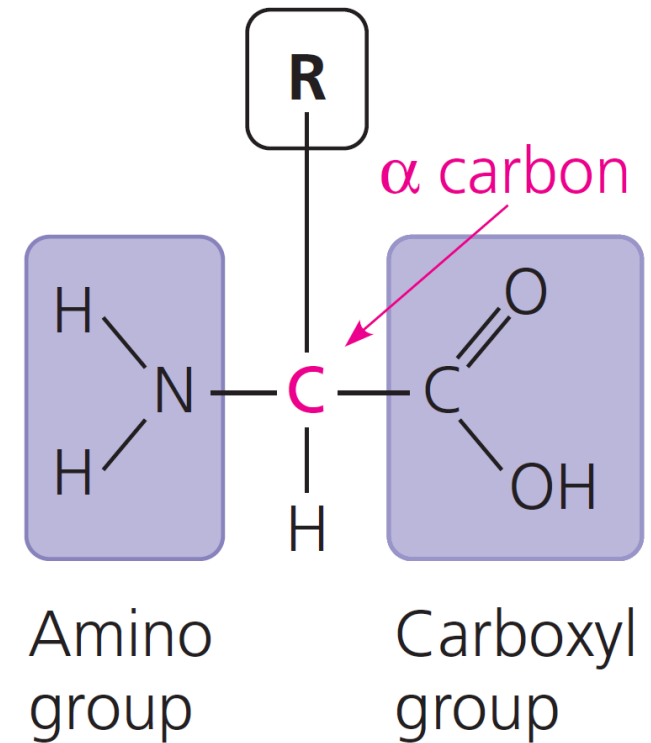
- 5.1 Macromolecules are polymers, built from monomers
- 5.2 Carbohydrates serve as fuel and building material
- 5.3 Lipids are a diverse group of hydrophobic molecules
- 5.4 Proteins include a diversity of structures, resulting in a wide range of functions
- 5.5 Nucleic acids store, transmit, and help express hereditary information
- 5.6 Genomics and proteomics have transformed biological inquiry and applications

Proteins



- ❖ Proteins account for **more than 50% of the dry mass of most cells**. They are instrumental in **almost everything an organism does**.
 - ✓ Protein functions include **structural support, storage, transport, cellular communication, movement, and defense against foreign substances**.
 - ✓ Most important, protein enzymes function as **catalysts** in cells, regulating metabolism by selectively accelerating certain chemical reactions without being consumed.
- ❖ Humans have **tens of thousands of different proteins**, each with a specific structure and function.
- ❖ Proteins are the **most structurally complex molecules known**.
 - ✓ Each type of protein has a **complex three-dimensional shape**.
 - ✓ All **proteins are unbranched polymers** constructed from the same 20 amino acid monomers.
 - ✓ Polymers of amino acids are called **polypeptides**.
 - ✓ A **protein** is a biologically functional molecule that consists of one or more polypeptides folded and coiled into a specific conformation.

Side chain (R group)

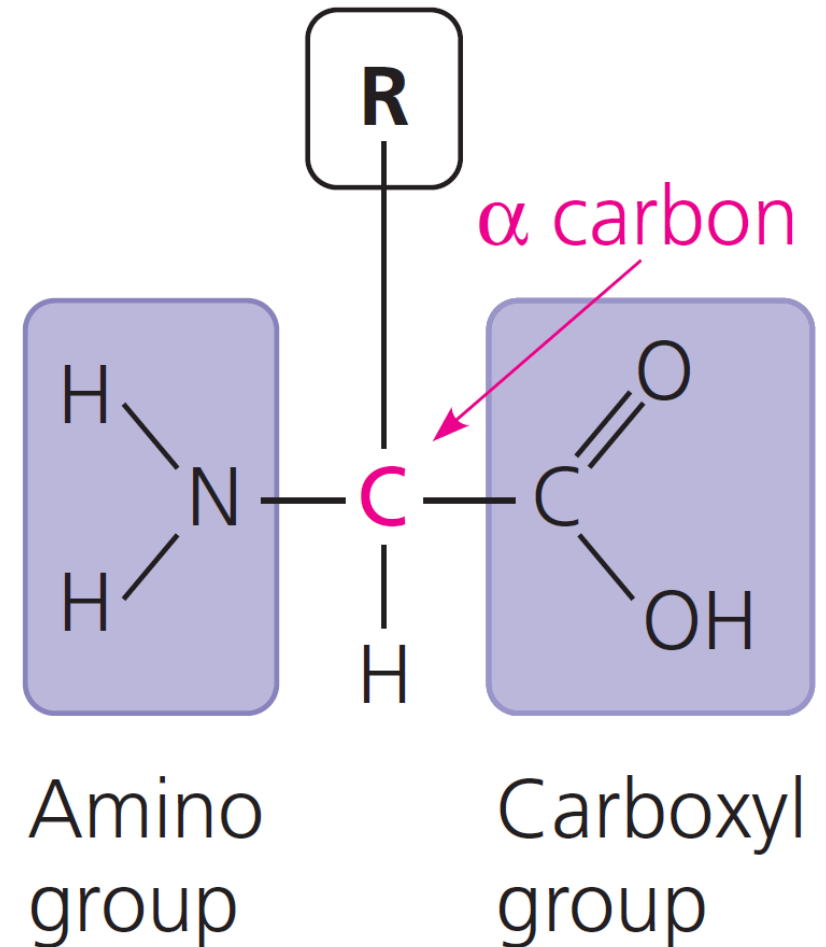


Proteins–structure and function



- ❖ **Amino acids** are organic molecules with both carboxyl and amino groups.
- ❖ At the center of an amino acid is an asymmetric carbon atom called the *alpha (α) carbon*.
- ❖ Four components are attached to the α carbon: a hydrogen atom, a carboxyl group, an amino group, and a variable **R group (or side chain)**.
 - ✓ *Different R groups characterize the 20 different amino acids.*
 - ✓ *An R group may be as simple as a hydrogen atom (as in the amino acid glycine), or it may be a carbon skeleton with various functional groups attached (as in glutamine).*

Side chain (R group)



Nature of R-groups

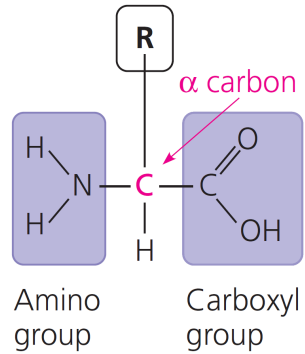


- ❖ The physical and chemical properties of the R group determine the unique characteristics of a particular amino acid.
 - ✓ *One group of amino acids has nonpolar R groups, which are **hydrophobic**.*
 - ✓ *Another group of amino acids has polar R groups, which are **hydrophilic**.*
 - ✓ *A third group of amino acids has functional groups that are **charged (ionized) at cellular pH**.*
 - ✓ *Some acidic R groups have **negative charge due to the presence of a carboxyl group**.*
 - ✓ *Basic R groups have **amino groups with positive charge**.*
 - ✓ *All amino acids have carboxyl and amino groups. The terms acidic and basic in this context refer only to these groups in the R groups.*

Nonpolar groups

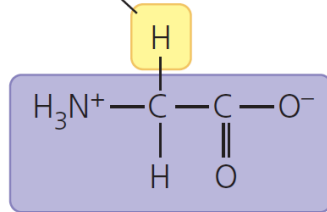


Side chain (R group)

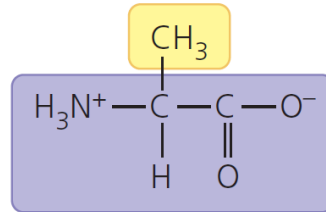


Nonpolar side chains; hydrophobic

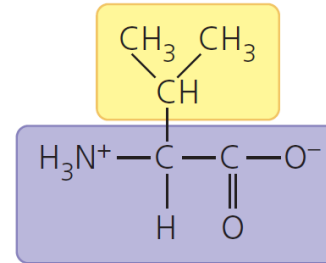
Side chain (R group)



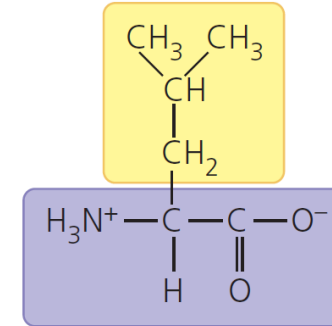
Glycine
(Gly or G)



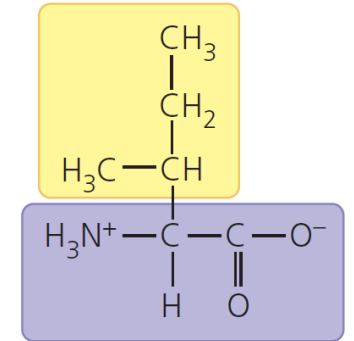
Alanine
(Ala or A)



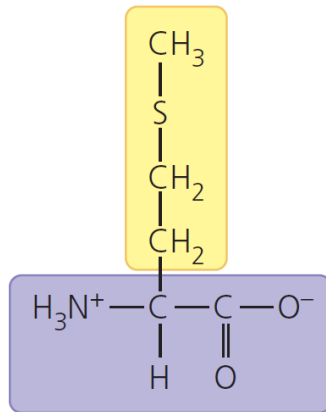
Valine
(Val or V)



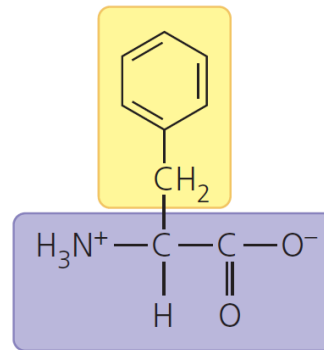
Leucine
(Leu or L)



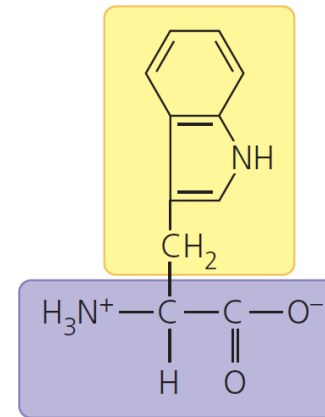
Isoleucine
(Ile or I)



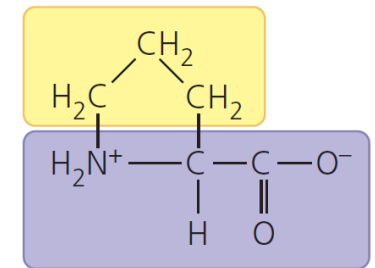
Methionine
(Met or M)



Phenylalanine
(Phe or F)



Tryptophan
(Trp or W)



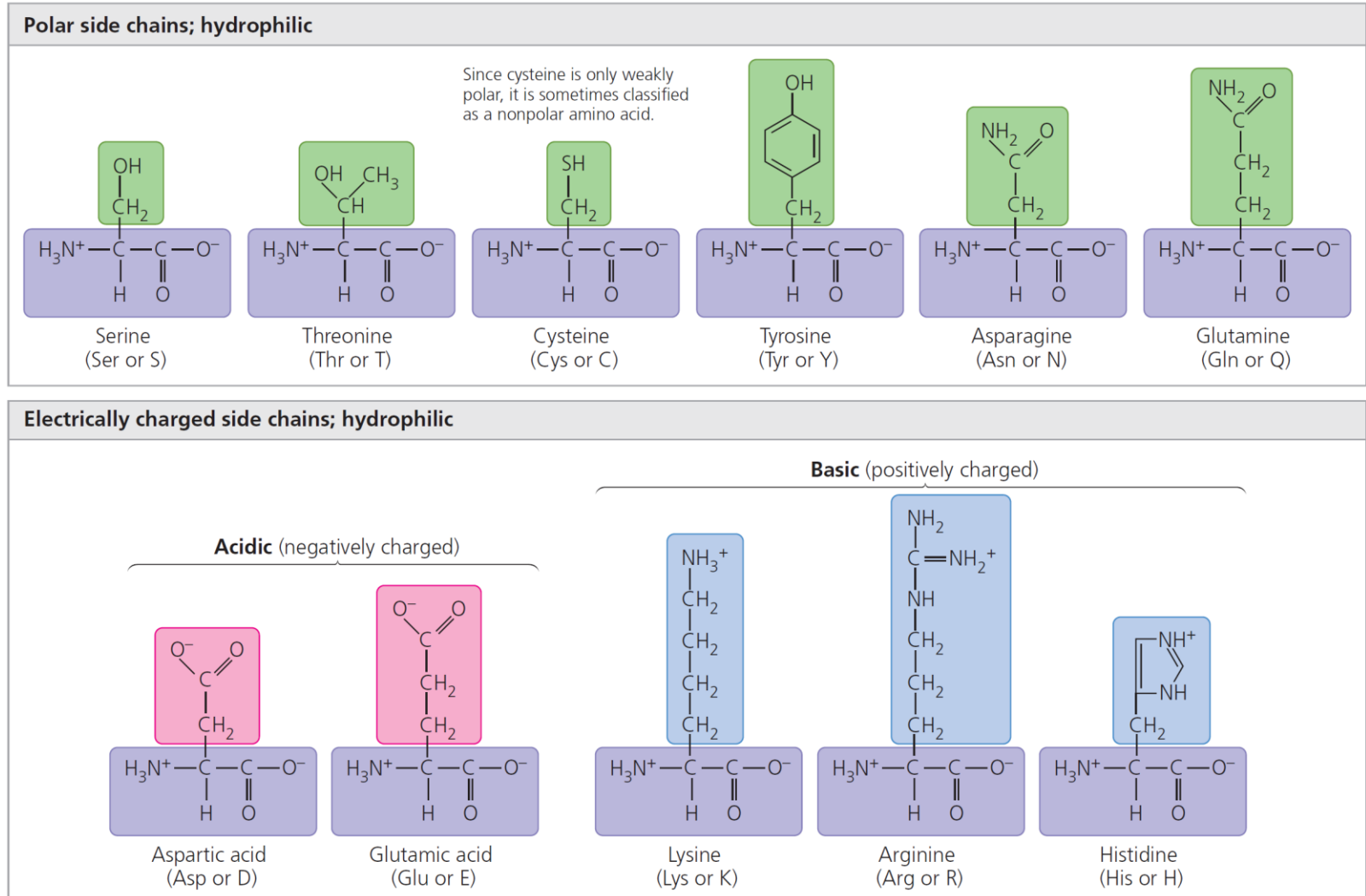
Proline
(Pro or P)

Amino acids have nonpolar R groups (shown in yellow shade), which are **hydrophobic**

Hydrophilic, electrically charged side groups



- ✓ Another group of amino acids has polar R groups, which are **hydrophilic**.
- ✓ Amino acids have functional groups that are **charged (ionized)** at cellular pH.
- ✓ Some acidic R groups have **negative charge** due to the presence of a carboxyl group.
- ✓ Basic R groups have **amino groups with positive charge**.



Peptides



- ❖ Amino acids are joined together when a dehydration reaction removes a hydroxyl group from the carboxyl end of one amino acid and a hydrogen atom from the amino group of another.

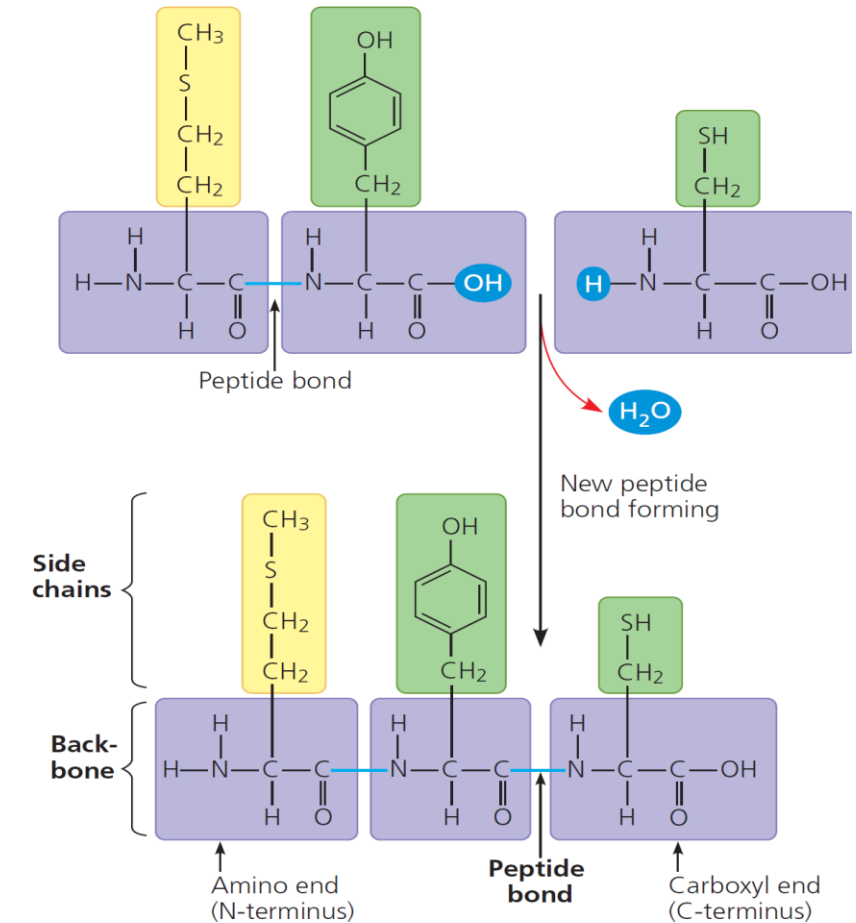
✓ *The resulting covalent bond is called a **peptide bond**.*

- ❖ Repeating the process over and over creates a polypeptide chain.

✓ *At one end is an amino acid with a free amino group (the **N-terminus**), and at the other end is an amino acid with a free carboxyl group (the **C-terminus**).*

- ❖ Polypeptides range in size from a few monomers to thousands.

- ❖ Each polypeptide has a unique linear sequence of amino acids.



▲ **Figure 5.15 Making a polypeptide chain.** Peptide bonds are formed by dehydration reactions, which link the carboxyl group of one amino acid to the amino group of the next. The peptide bonds are formed one at a time, starting with the amino acid at the amino end (N-terminus). The polypeptide has a repetitive backbone (purple) to which the amino acid side chains (yellow and green) are attached.

Amino Acid	Three-Letter Abbreviation	One-Letter Symbol	Molecular Weight
Alanine	Ala	A	89 Da
Arginine	Arg	R	174 Da
Asparagine	Asn	N	132 Da
Aspartic acid	Asp	D	133 Da
Asparagine or aspartic acid	Asx	B	133 Da
Cysteine	Cys	C	121 Da
Glutamine	Gln	Q	146 Da
Glutamic acid	Glu	E	147 Da
Glutamine or glutamic acid	Glx	Z	147 Da
Glycine	Gly	G	75 Da
Histidine	His	H	155 Da
Isoleucine	Ile	I	131 Da
Leucine	Leu	L	131 Da
Lysine	Lys	K	146 Da
Methionine	Met	M	149 Da
Phenylalanine	Phe	F	165 Da
Proline	Pro	P	115 Da
Serine	Ser	S	105 Da
Threonine	Thr	T	119 Da
Tryptophan	Trp	W	204 Da
Tyrosine	Tyr	Y	181 Da
Valine	Val	V	117 Da

- ❖ From the table the **average molecular weight of an amino acid is 110 Da.**
 - ❖ Dalton (Da) is an alternate name for the atomic mass unit, and kilodalton (kDa) is 1,000 daltons.
 - ❖ Thus a protein with a mass of 64 kDa has a molecular weight of 64,000 grams per mole (g/mole).
 - ❖ M.W. of protein =

$$\# \text{ amino acids} \times 110 \text{ Da}$$
[Rule of thumb!]
- (Note: **BJOUXZ** alphabets are not part of 20 naturally occurring amino acids.)

Protein functions–variety endless

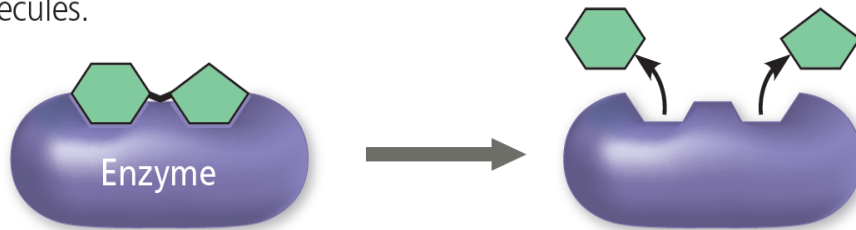


Protein conformation determines protein function.

Enzymatic proteins

Function: Selective acceleration of chemical reactions

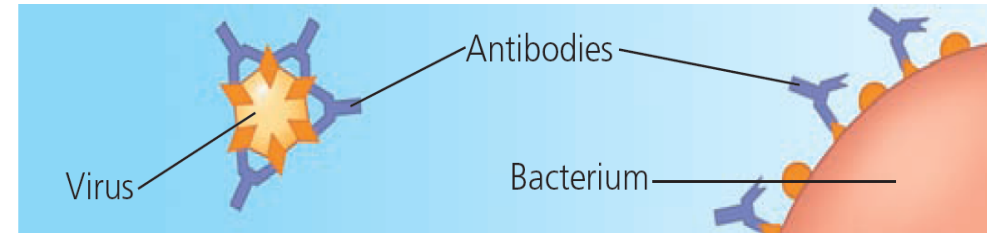
Example: Digestive enzymes catalyze the hydrolysis of bonds in food molecules.



Defensive proteins

Function: Protection against disease

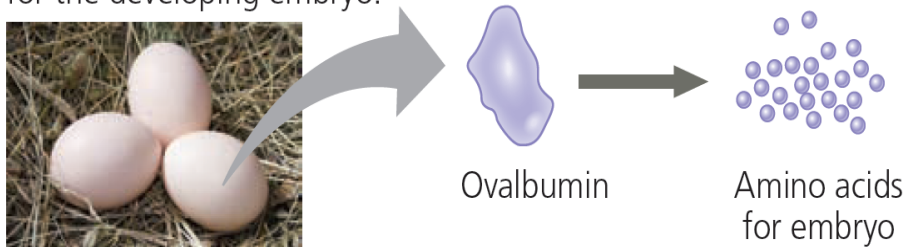
Example: Antibodies inactivate and help destroy viruses and bacteria.



Storage proteins

Function: Storage of amino acids

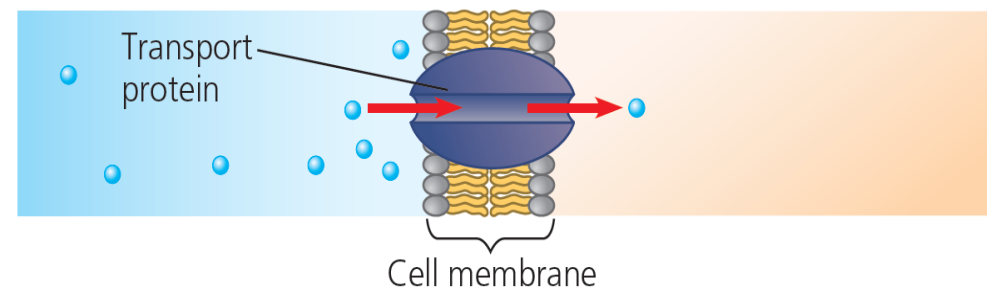
Examples: Casein, the protein of milk, is the major source of amino acids for baby mammals. Plants have storage proteins in their seeds. Ovalbumin is the protein of egg white, used as an amino acid source for the developing embryo.



Transport proteins

Function: Transport of substances

Examples: Hemoglobin, the iron-containing protein of vertebrate blood, transports oxygen from the lungs to other parts of the body. Other proteins transport molecules across membranes, as shown here.



Protein functions–variety endless

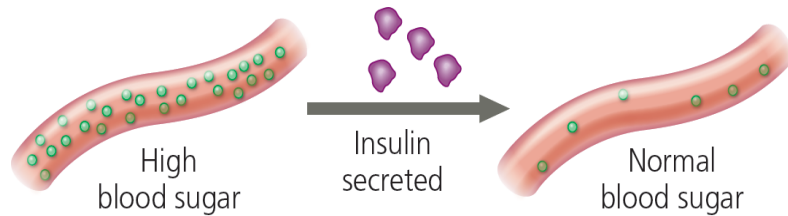


Protein conformation determines protein function.

Hormonal proteins

Function: Coordination of an organism's activities

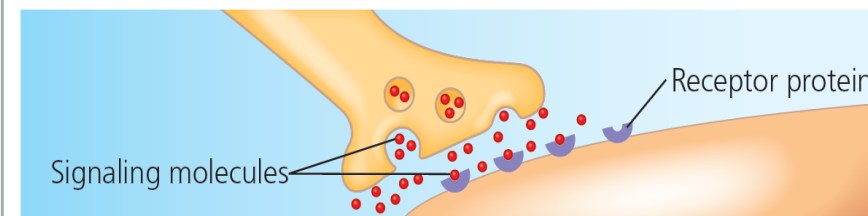
Example: Insulin, a hormone secreted by the pancreas, causes other tissues to take up glucose, thus regulating blood sugar concentration.



Receptor proteins

Function: Response of cell to chemical stimuli

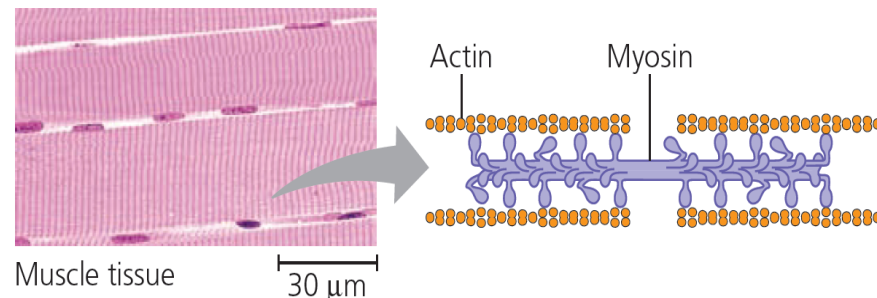
Example: Receptors built into the membrane of a nerve cell detect signaling molecules released by other nerve cells.



Contractile and motor proteins

Function: Movement

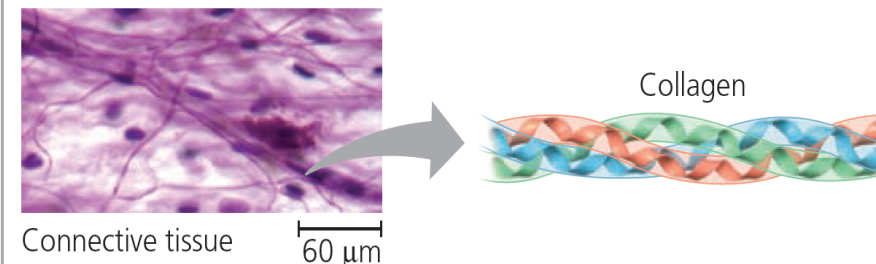
Examples: Motor proteins are responsible for the undulations of cilia and flagella. Actin and myosin proteins are responsible for the contraction of muscles.



Structural proteins

Function: Support

Examples: Keratin is the protein of hair, horns, feathers, and other skin appendages. Insects and spiders use silk fibers to make their cocoons and webs, respectively. Collagen and elastin proteins provide a fibrous framework in animal connective tissues.



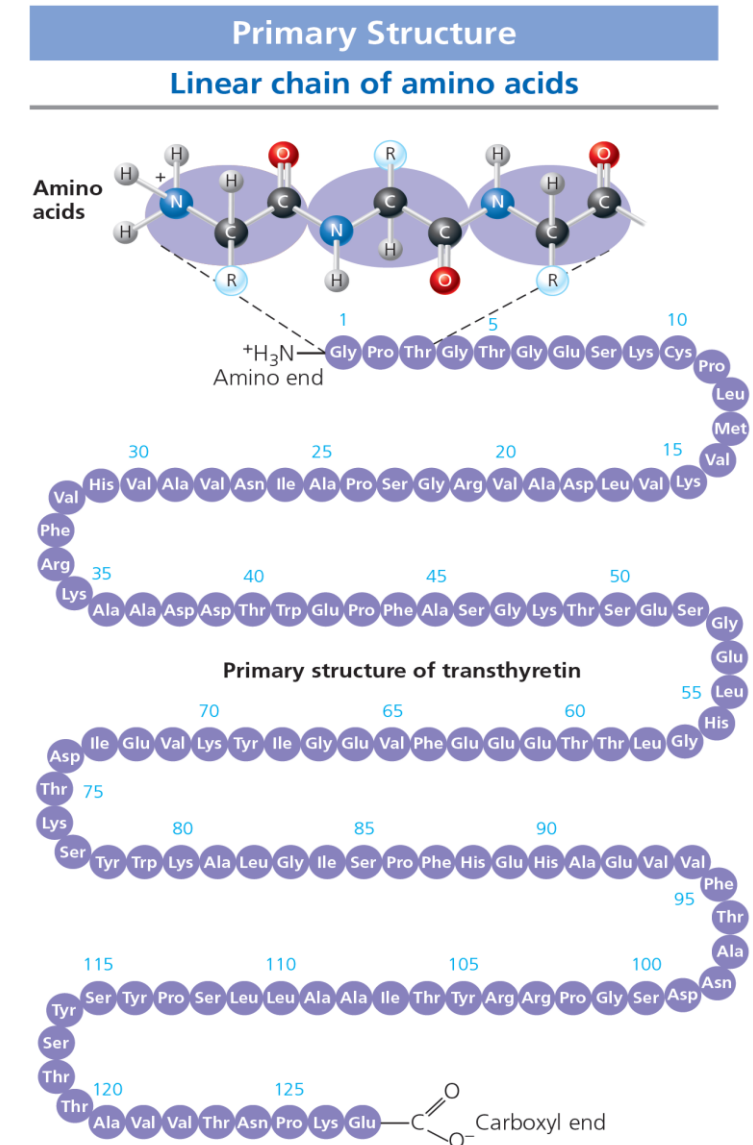
Proteins–primary structure



❖ The **primary structure** of a protein is its unique sequence of amino acids.

❖ Transthyretin is a globular protein found in the blood that transports vitamin A and a particular thyroid hormone throughout the body.

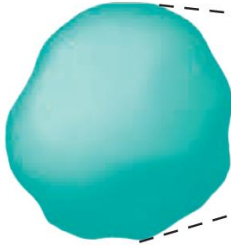
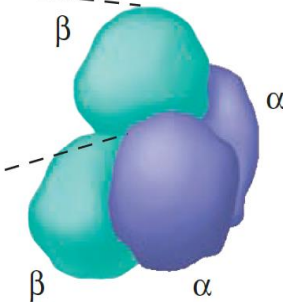
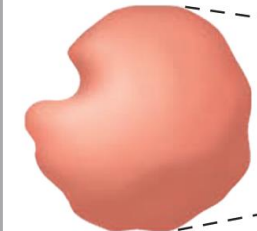
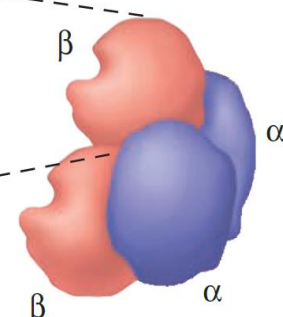
- ✓ *Each of the four identical polypeptide chains that, together, make up transthyretin is composed of 127 amino acids.*
- ✓ *The precise primary structure of each of the polypeptide chains is determined by **inherited genetic information**, not by the random linking of amino acids.*



Protein-secondary structures



- ❖ Most proteins have segments of their polypeptide chains **repeatedly coiled or folded**.
- ❖ These coils and folds are referred to as **secondary structure** and result from hydrogen bonds between the repeating constituents of the polypeptide backbone.
 - ✓ *The weakly positive hydrogen atom attached to the nitrogen atom has an affinity for the oxygen atom of a nearby peptide bond.*
 - ✓ *Each hydrogen bond is weak, but the sum of many hydrogen bonds stabilizes the structure of part of the protein.*
- ❖ One secondary structure is the **α helix**, a delicate coil held together by hydrogen bonding between every fourth amino acid, as found in transthyretin.
 - ✓ *Some fibrous proteins, such as α -keratin, the structural protein of hair, have the α helix formation over most of their length.*

	Primary Structure	Secondary and Tertiary Structures	Quaternary Structure
Normal hemoglobin	1 Val 2 His 3 Leu 4 Thr 5 Pro 6 Glu 7 Glu	Normal β subunit 	Normal hemoglobin 
Sickle-cell hemoglobin	1 Val 2 His 3 Leu 4 Thr 5 Pro 6 Val 7 Glu	Sickle-cell β subunit 	Sickle-cell hemoglobin 

Secondary and tertiary structures

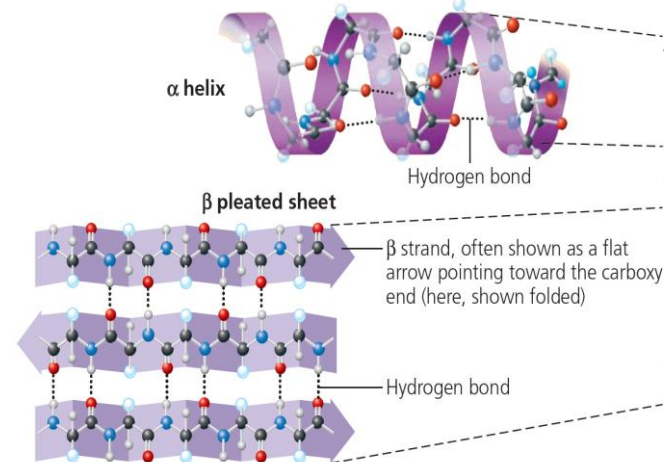


❖ The other main type of secondary structure is the **β pleated sheet**.

- ✓ In this structure, two or more regions of the polypeptide chain lying side by side are connected by hydrogen bonds between parts of the two parallel polypeptide backbones.
- ✓ Pleated sheets are found in many globular proteins, such as transthyretin, and they dominate some fibrous proteins, including the silk protein of a spider's web.
- ✓ The presence of so many hydrogen bonds makes each silk fiber stronger than a steel strand of the same weight.

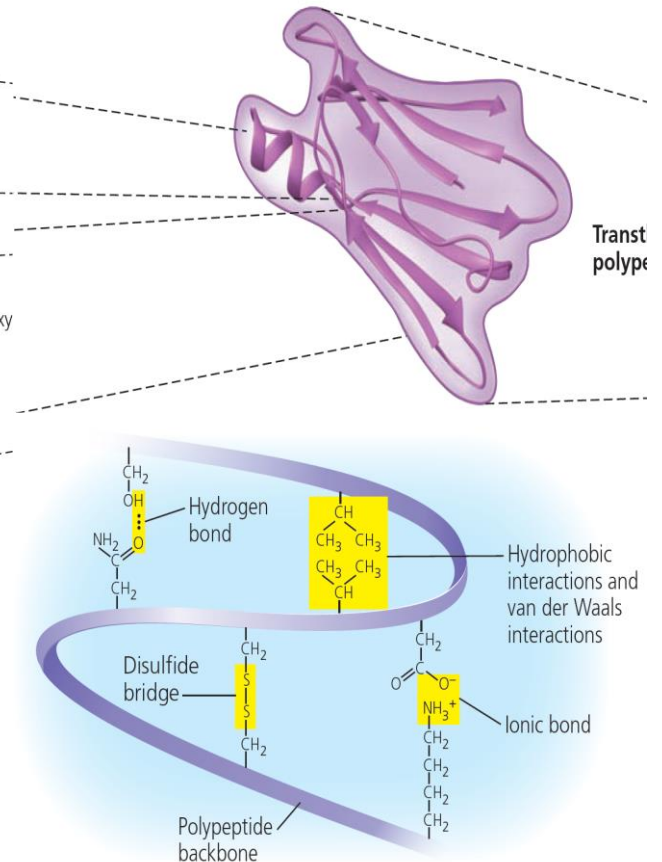
Secondary Structure

Regions stabilized by hydrogen bonds between atoms of the polypeptide backbone



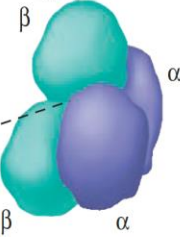
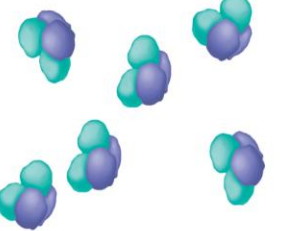
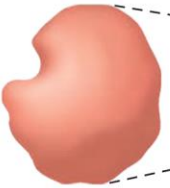
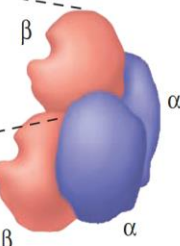
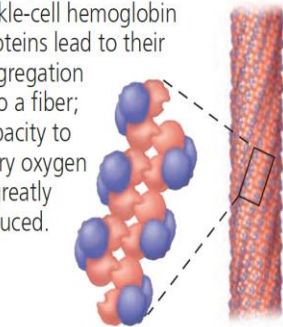
Tertiary Structure

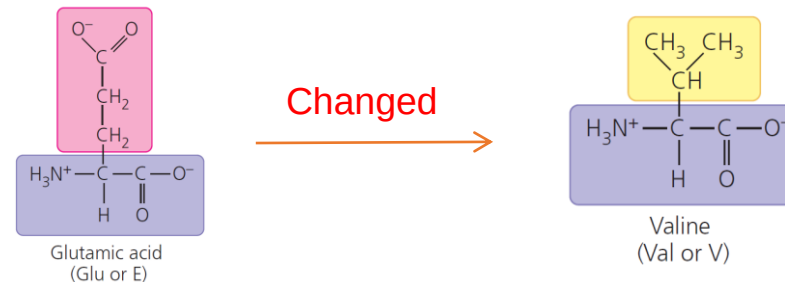
Three-dimensional shape stabilized by interactions between side chains



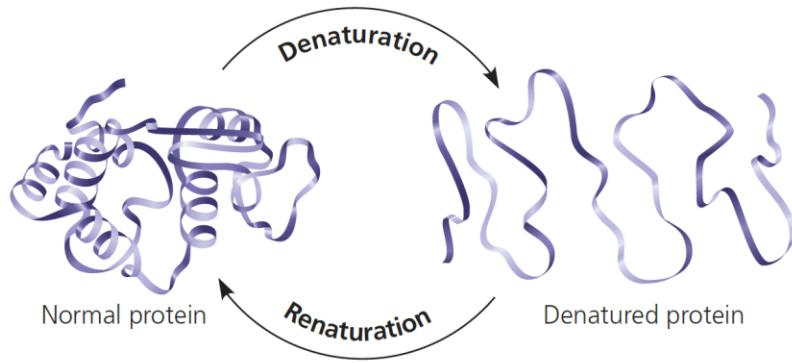
Malfunction of protein structures–consequences

- ❖ Even a slight change in the primary structure can affect a protein's conformation and ability to function.
- ✓ *The substitution of one amino acid (valine) for the normal one (glutamic acid) at a particular position in the primary structure of hemoglobin, the protein that carries oxygen in red blood cells, can cause **sickle-cell disease**, an inherited blood disorder.*
- ✓ *The **abnormal hemoglobin** molecules crystallize, deforming some of the red blood cells into a sickle shape and clogging capillaries.*

	Primary Structure	Secondary and Tertiary Structures	Quaternary Structure	Function	Red Blood Cell Shape
Normal hemoglobin	1 Val 2 His 3 Leu 4 Thr 5 Pro 6 Glu 7 Glu	Normal β subunit 	Normal hemoglobin 	Normal hemoglobin proteins do not associate with one another; each carries oxygen. 	Normal red blood cells are full of individual hemoglobin proteins.  5 μ m
Sickle-cell hemoglobin	1 Val 2 His 3 Leu 4 Thr 5 Pro 6 Val 7 Glu	Sickle-cell β subunit 	Sickle-cell hemoglobin 	Hydrophobic interactions between sickle-cell hemoglobin proteins lead to their aggregation into a fiber; capacity to carry oxygen is greatly reduced. 	Fibers of abnormal hemoglobin deform red blood cell into sickle shape.  5 μ m



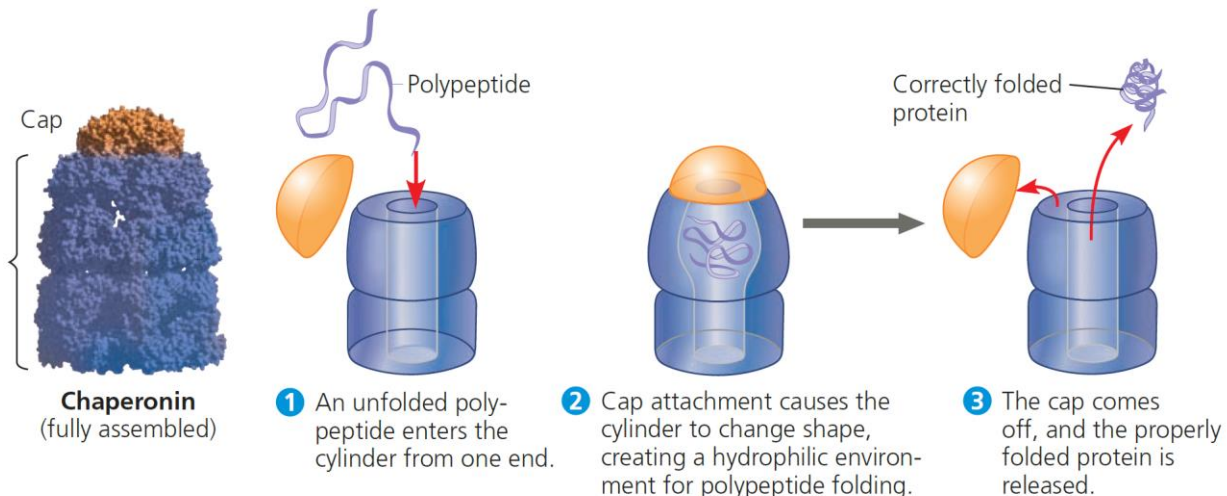
Protein folding-chaperones



▲ **Figure 5.20 Denaturation and renaturation of a protein.** High temperatures or various chemical treatments will denature a protein, causing it to lose its shape and hence its ability to function. If the denatured protein remains dissolved, it may renature when the chemical and physical aspects of its environment are restored to normal.

- ❖ The folding of many proteins is assisted by **chaperonins**, or chaperone proteins.
 - ✓ *Chaperonins do not specify the final structure of a polypeptide but rather work to segregate and protect the polypeptide while it folds spontaneously.*
 - ✓ *Molecular systems in the cell interact with chaperonins, marking incorrectly folded proteins for refolding or for destruction.*
- ❖ Accumulation of **incorrectly folded polypeptides** is associated with many diseases, including **Alzheimer's, Parkinson's, and mad cow disease**.

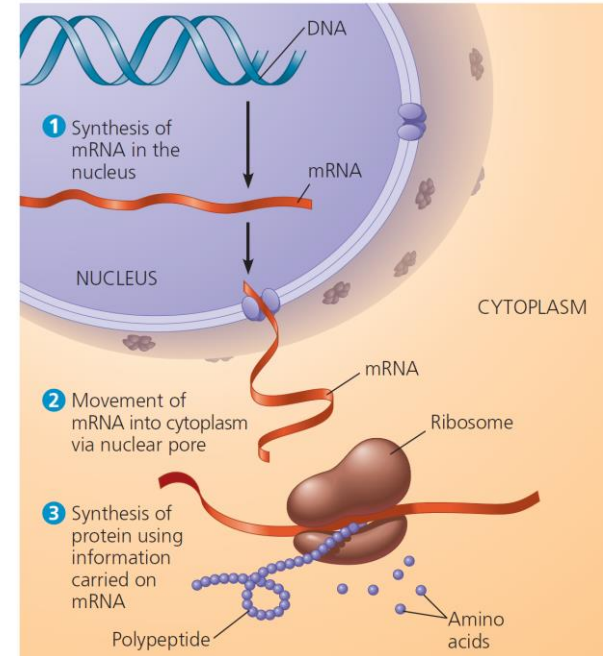
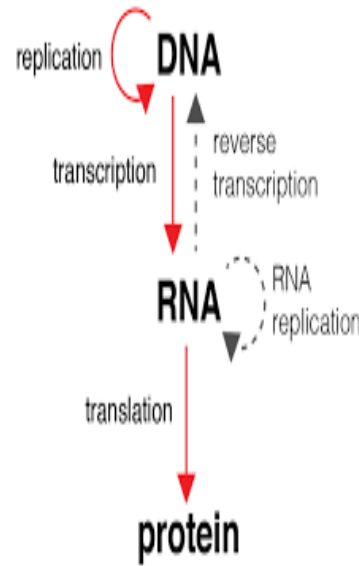
► **Figure 5.21 A chaperonin in action.** The computer graphic (left) shows a large chaperonin protein complex from the bacterium *E. coli*. It has an interior space that provides a shelter for the proper folding of newly made polypeptides. The complex consists of two proteins: One is a hollow cylinder; the other is a cap that can fit on either end. The steps of chaperonin activity are shown at the right.



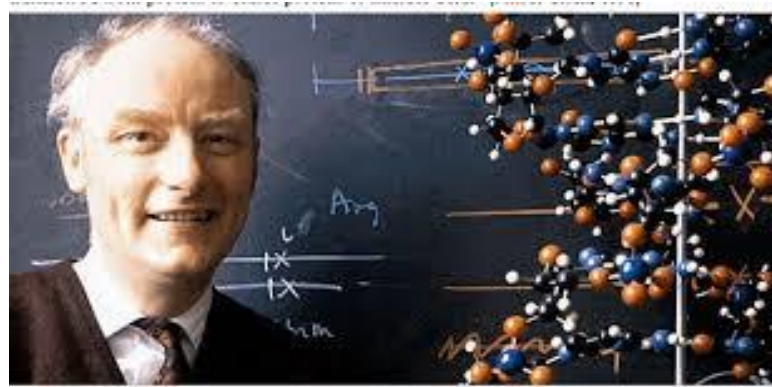
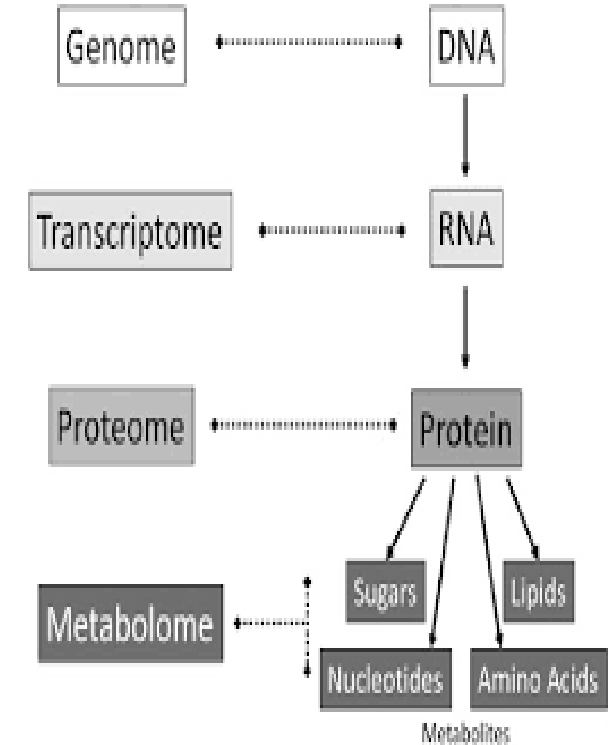
Nucleic acids–central dogma



- ❖ The amino acid sequence of a polypeptide is programmed by a discrete unit of inheritance known as a **gene**.
- ❖ A gene consists of DNA, a polymer known as a **nucleic acid**.
 - ✓ Nucleic acids are made of monomers called *nucleotides*.
- ❖ The two types of nucleic acids are
 - ✓ **ribonucleic acid (RNA)** and
 - ✓ **deoxyribonucleic acid (DNA)**.
- ❖ RNA and DNA are the molecules that enable living organisms to reproduce their complex components from generation to generation.



▲ **Figure 5.23** Gene expression: DNA → RNA → protein. In a eukaryotic cell, DNA in the nucleus programs protein production in the cytoplasm by dictating synthesis of messenger RNA (mRNA).



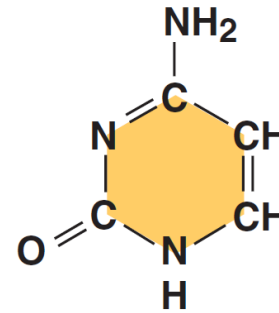
- ❖ The '**Central Dogma**' is the process by which the instructions in DNA are converted into a functional product.
- ❖ It was first proposed in 1958 by Francis Crick, discoverer of the structure of DNA.

Nucleic acids– Nitrogenous bases

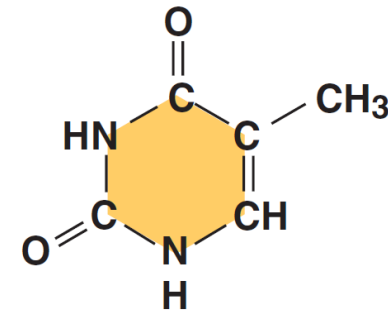


- ❖ Nucleic acids are polymers made of **nucleotide** monomers organized as **polynucleotides**.
- ❖ Each **nucleotide** consists of three parts: a nitrogenous base, a pentose sugar, and one or more phosphate groups.
- ❖ The nitrogenous bases are rings of carbon and nitrogen that come in two types: purines and pyrimidines.
 - **Pyrimidines** have a single six-membered ring of carbon and nitrogen atoms.
 - ✓ There are **three different pyrimidines**: cytosine (C), thymine (T), and uracil (U).
 - ✓ Thymine is found only in DNA and uracil is found only in RNA.
 - **Purines** have a six-membered ring joined to a five-membered ring.
 - ✓ The two purines are adenine (A) and guanine (G).

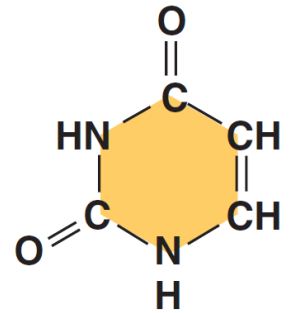
Pyrimidines



Cytosine (C)

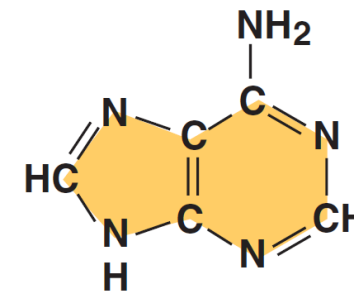


Thymine (T, in DNA)

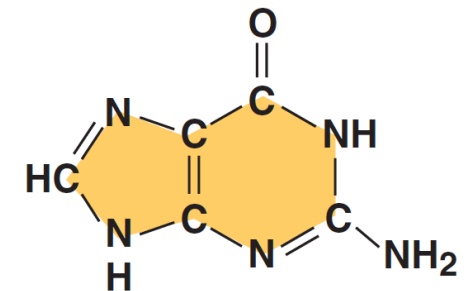


Uracil (U, in RNA)

Purines



Adenine (A)

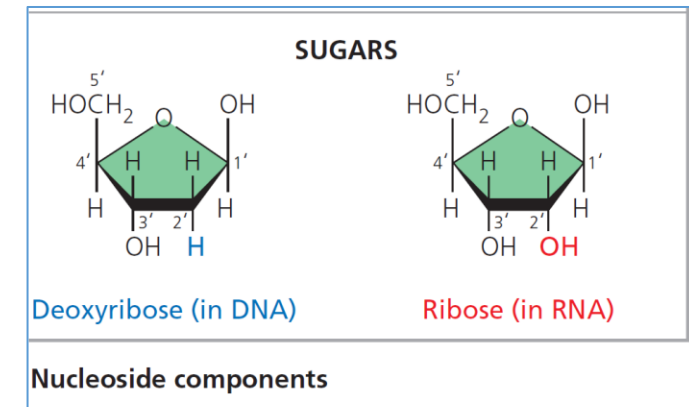


Guanine (G)

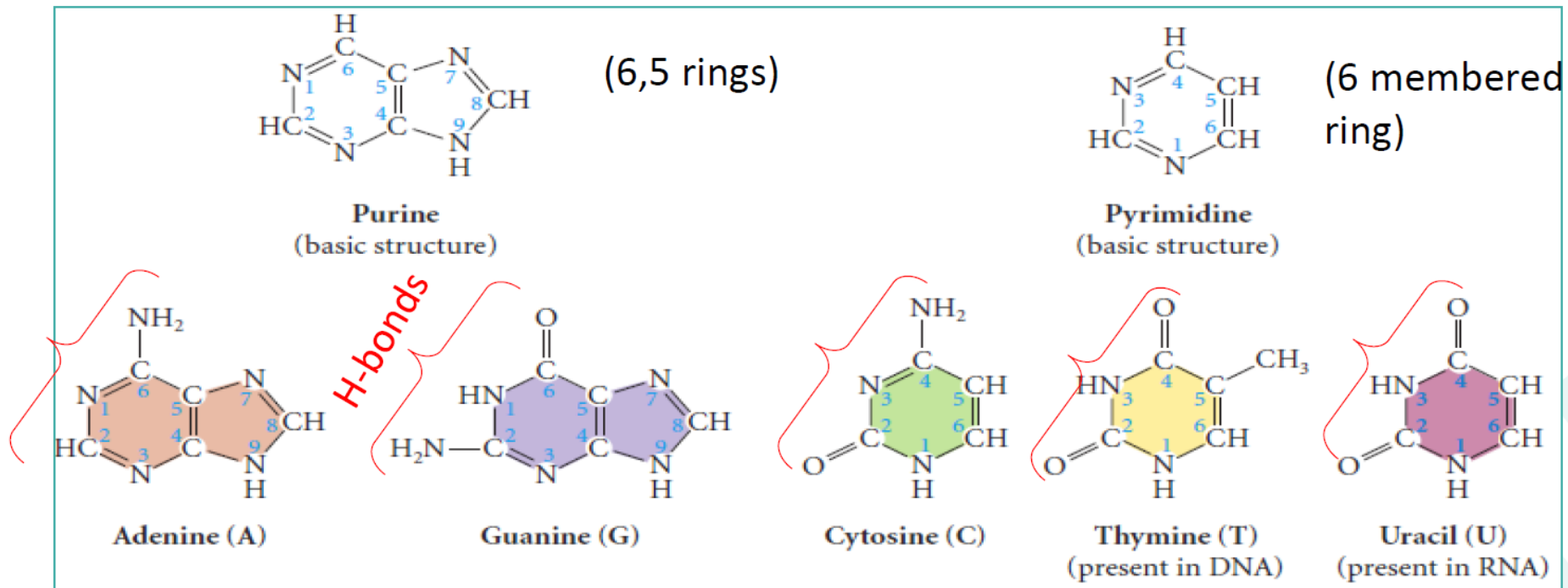
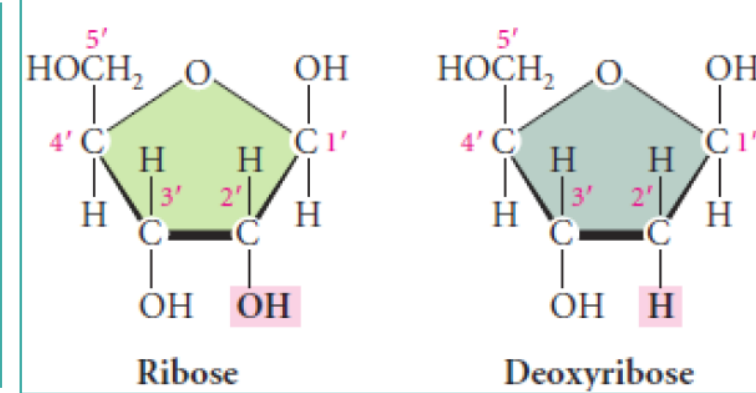
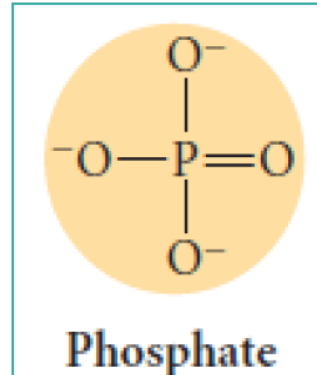
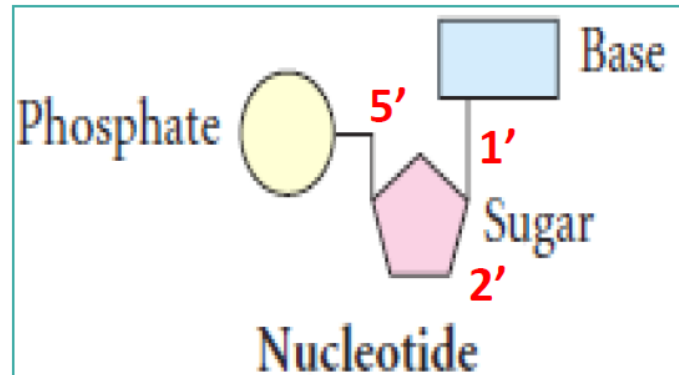
Nucleic acid – sugar component



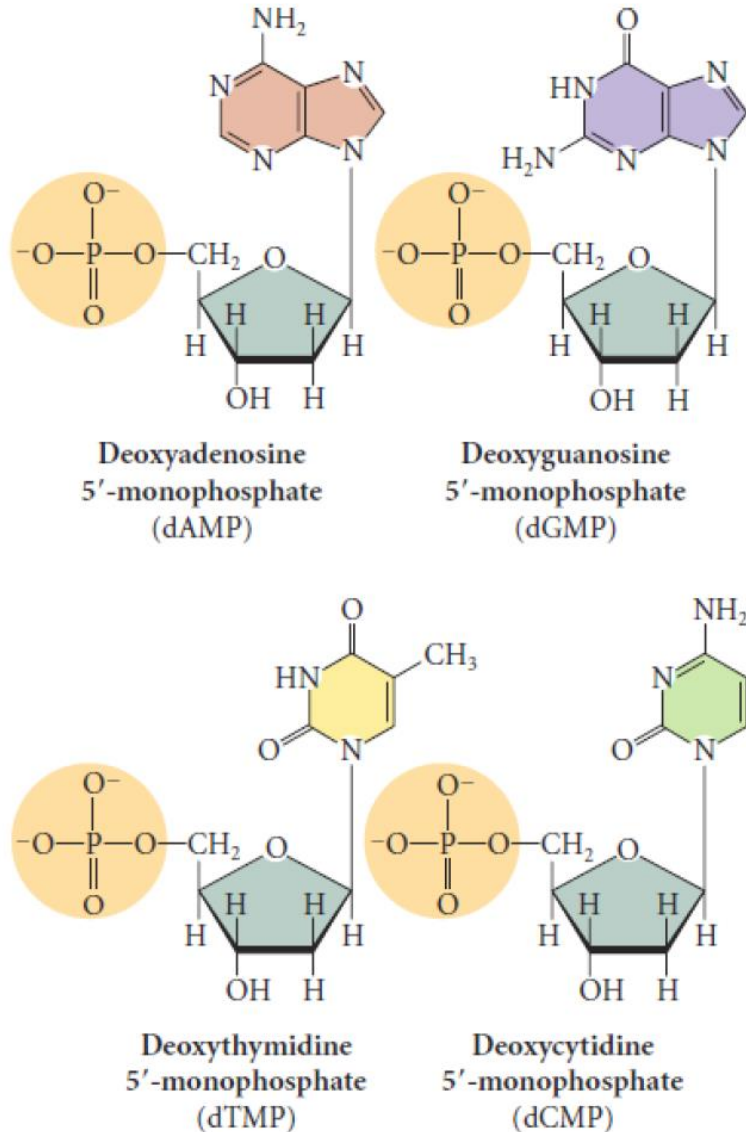
- ❖ The pentose joined to the nitrogenous base is **ribose** in nucleotides of RNA and **deoxyribose** in DNA.
 - ✓ *The only difference between the sugars is the lack of an oxygen atom on carbon 2 in deoxyribose.*
 - ✓ *Because the atoms in both the nitrogenous base and the sugar are numbered, the sugar atoms are distinguished by a prime (') after the number.*
 - ✓ *Thus, the second carbon in the sugar ring is the 2' (2 prime) carbon, and the carbon that sticks up from the ring is the 5' carbon.*
 - ✓ *The combination of a pentose and a nitrogenous base is a **nucleoside**.*
- ❖ The addition of a phosphate group creates a nucleoside monophosphate or *nucleotide*.
- ❖ **Polynucleotides are synthesized when adjacent nucleotides are joined by covalent bonds called phosphodiester linkages** that form between the —OH group on the 3' of one nucleotide and the phosphate on the 5' carbon of the next.
 - ✓ This process creates a repeating backbone of sugar-phosphate units, with appendages consisting of the nitrogenous bases.



Summary: Structure of nucleotide, nucleosides, ribose and DNA



Four types of DNA nucleotides



CONCEPT CHECK

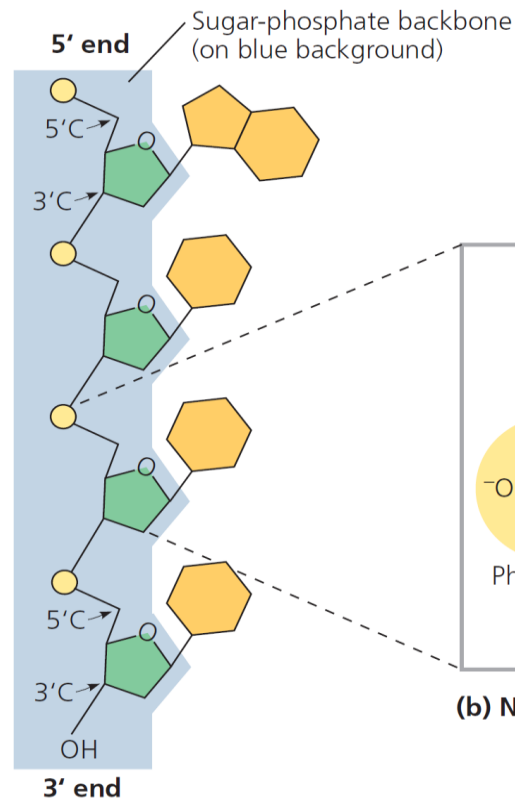
Q: How do the sugars of RNA and DNA differ?

- a. RNA has a six-carbon sugar; DNA has a five-carbon sugar.
- b. The sugar of RNA has a hydroxyl group that is not found in the sugar of DNA.
- c. RNA contains uracil; DNA contains thymine.
- d. DNA's sugar has a phosphorus atom; RNA's sugar does not.

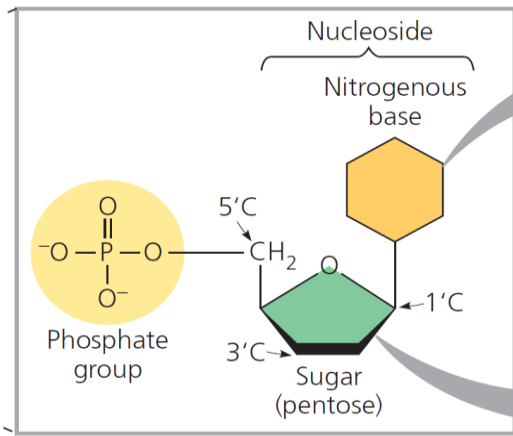
Nucleic acids – reads from 5' → 3'



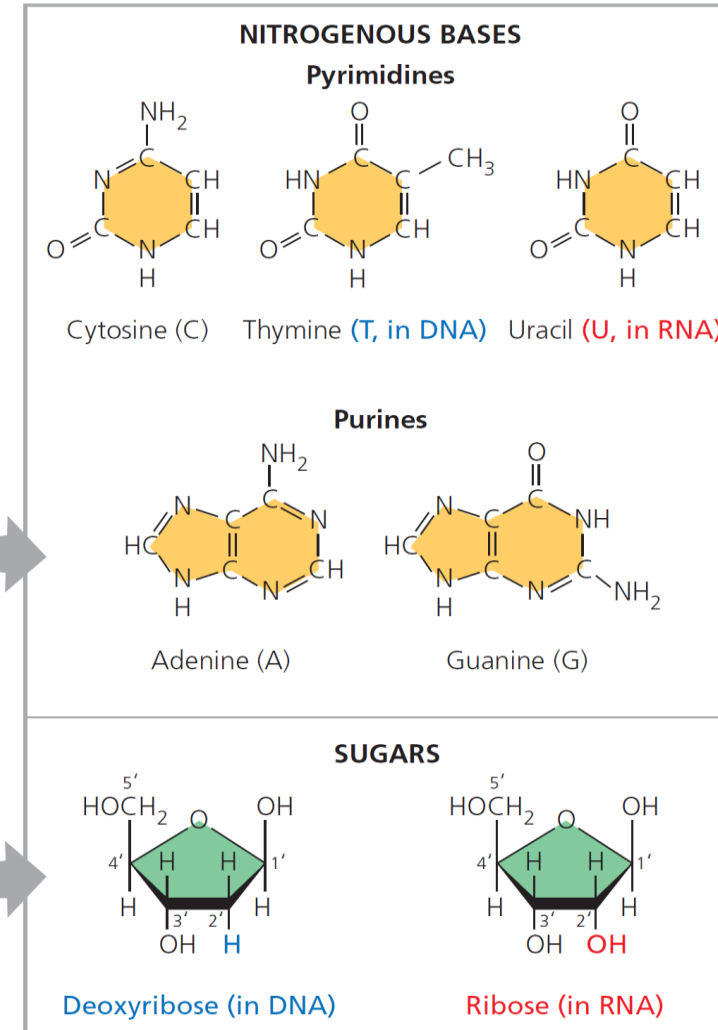
▼ **Figure 5.24 Components of nucleic acids.** (a) A polynucleotide has a sugar-phosphate backbone with variable appendages, the nitrogenous bases. (b) A nucleotide monomer includes a nitrogenous base, a sugar, and a phosphate group. Note that carbon numbers in the sugar include primes ('). (c) A nucleoside includes a nitrogenous base (purine or pyrimidine) and a five-carbon sugar (deoxyribose or ribose).



(a) Polynucleotide, or nucleic acid

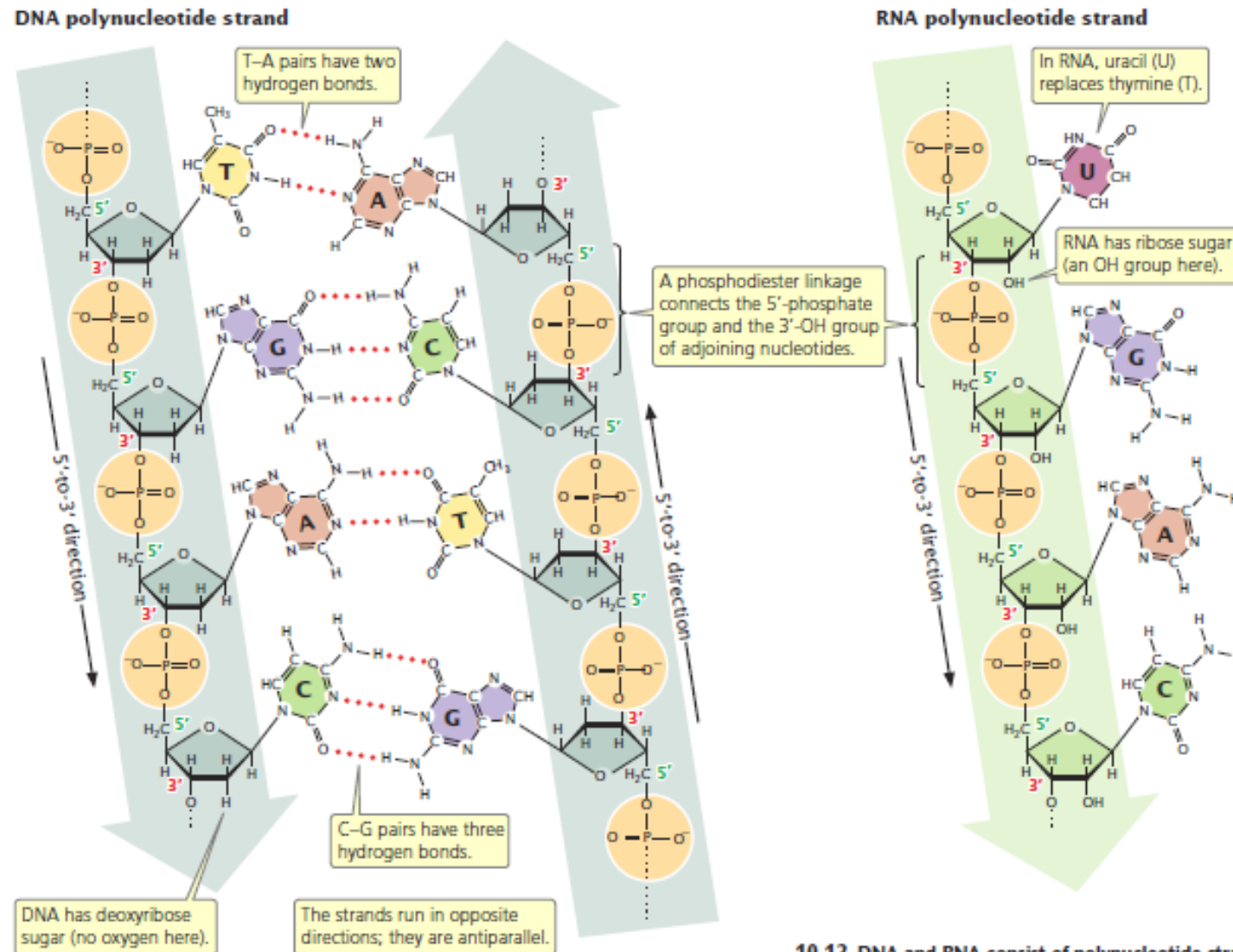


(b) Nucleotide



(c) Nucleoside components

Strands in DNA—5' is phosphate end and 3' is OH end of sugar

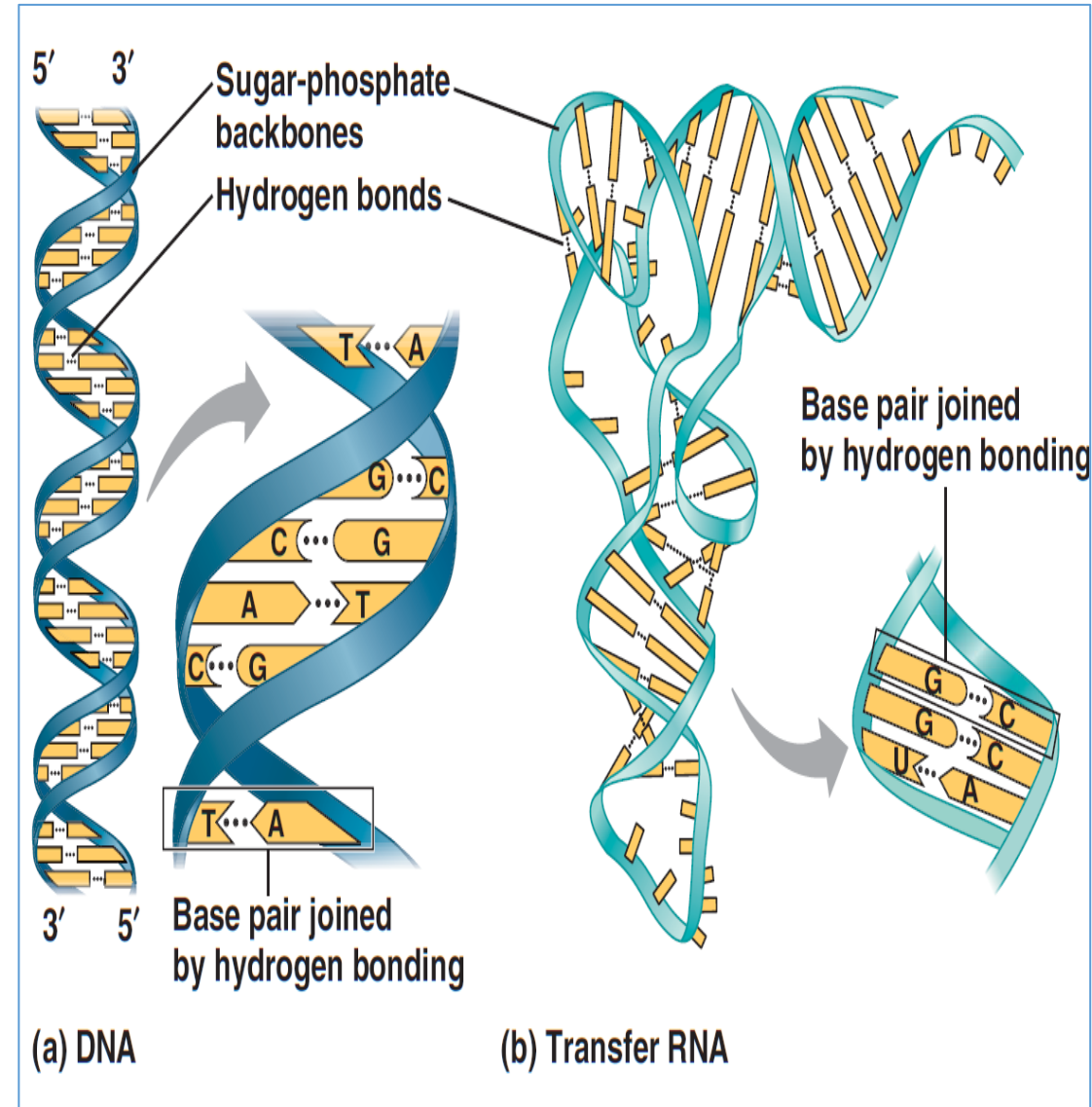


10.13 DNA and RNA consist of polynucleotide strands.

DNA-hydrogen bonding



- ❖ An **RNA** molecule is **usually** a **single polynucleotide chain**.
- ❖ DNA molecules have two polynucleotide strands that spiral around an imaginary axis to form a **double helix**.
- ❖ The sugar-phosphate backbones of the two polynucleotides are on the outside of the helix.
 - ✓ *The two backbones run in opposite 5' → 3' directions from each other, an arrangement referred to as **antiparallel**.*
- ❖ The two polynucleotides or strands are held together by hydrogen bonds between the paired bases.
- ❖ Most DNA molecules have thousands to millions of base pairs.
- ❖ Because of their shapes, **only some bases are compatible** with each other.
 - ✓ *Adenine (A) always pairs with thymine (T) and guanine (G) with cytosine (C).*
 - ✓ *With these base-pairing rules, **if we know the sequence of bases on one strand, we know the sequence on the opposite strand.***
 - ✓ *The two strands are **complementary**.*



Applications

in Science and Technology



- ❖ Nucleotide sequencing and the analysis of large sets of genes and proteins can be done rapidly and inexpensively due to advances in technology and information processing.
- ❖ Taken together, **genomics and proteomics** have advanced our understanding of biology across many different fields.

Paleontology

New DNA sequencing techniques have allowed decoding of minute quantities of DNA found in ancient tissues from our extinct relatives, the Neanderthals (*Homo neanderthalensis*). Sequencing the Neanderthal genome has informed our understanding of their physical appearance as well as their relationship with modern humans. See Figure 34.49.



Evolution

A major aim of evolutionary biology is to understand the relationships among species, both living and extinct. For example, genome sequence comparisons have identified the hippopotamus as the land mammal sharing the most recent common ancestor with whales. See Figure 22.20.



Hippopotamus

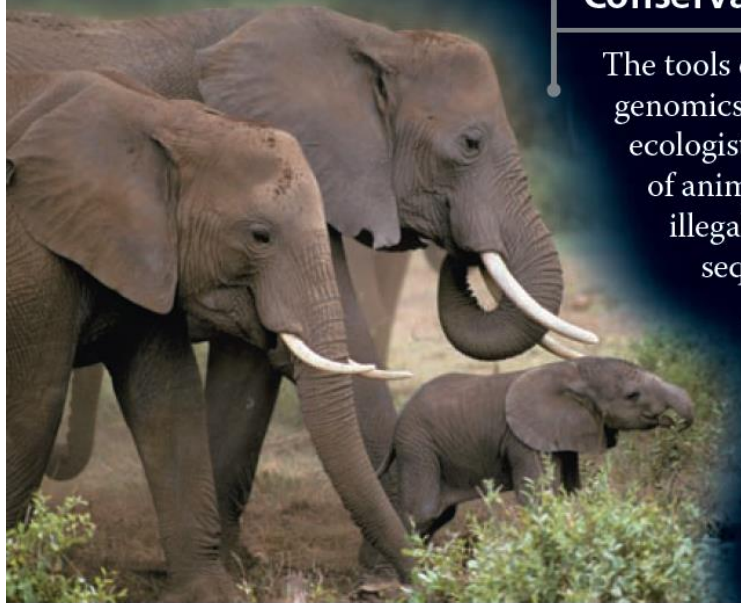
Medical Science

Identifying the genetic basis for human diseases like cancer helps researchers focus their search for potential future treatments. Currently, sequencing the sets of genes expressed in an individual's tumor can allow a more targeted approach to treating the cancer, a type of "personalized medicine." *See Figures 12.20 and 18.27.*



Conservation Biology

The tools of molecular genetics and genomics are increasingly used by ecologists to identify which species of animals and plants are killed illegally. In one case, genomic sequences of DNA from illegal shipments of elephant tusks were used to track down poachers and pinpoint the territory where they were operating. *See Figure 56.9.*



Species Interactions

Over 90% of all plant species exist in a mutually beneficial partnership with fungi that are associated with the plants' roots. Genome sequencing and analysis of gene expression in several plant-fungal pairs promise major advances in our understanding of such interactions and may have implications for agricultural practices. *(See the Scientific Skills Exercise in Chapter 31.)*



MAKE CONNECTIONS Considering the examples provided here, describe how the approaches of genomics and proteomics help us to address a variety of biological questions.